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Performance Analysis and Enhancements of Memory Systems for Multi-Chiplet NUMA Architectures

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Abstract

As the semiconductor industry is struggling with the slowdown in Moore's Law and the challenges of increased design complexity on a single chip, multi-chiplet systems have become a promising alternative to large monolithic systems, offering improved yields and lower costs. However, these chiplet-based architectures are susceptible to non-uniform memory access (NUMA) inefficiencies, where remote memory access significantly hinders the system's performance. These performance bottlenecks are augmented by the high latency and limited bandwidth of the inter-chiplet communication, thus compromising the overall performance of multi-chiplet systems. While prior studies have thoroughly examined the yield and cost benefits of multi-chiplet chips, their performance relative to monolithic counterparts remains unexplored. This thesis delves into a comprehensive performance analysis of multi-chiplet systems, comparing them to traditional monolithic designs and evaluating their cost-performance trade-offs. While multi-chiplet systems can drastically reduce recurring engineering costs by nearly half, our analysis reveals that they may suffer performance losses of up to one-third compared to monolithic systems due to these NUMA-related overheads. To address the performance overheads, this thesis introduces MEMPLEX, a novel memory system explicitly designed for multi-chiplet NUMA architectures. MEMPLEX combines data replication and migration strategies to optimize data placement and improve data locality within the multi-chiplet memory hierarchy. By allocating a portion of each memory node as a DRAM cache and enabling migration based on access patterns and memory traffic, MEMPLEX reduces the frequency of costly remote memory accesses, mitigates performance overheads, and delivers substantial energy savings. The evaluation on multi-programmed workloads from different benchmark suites demonstrated that, compared to a multi-chiplet system with NUMA-aware data placement and no support for DRAM caching or migration, MEMPLEX reduces remote memory traffic by 80%, leading to a significant 44% dynamic memory energy consumption. MEMPLEX also delivers up to 7% speedup (5% on average) when $\frac{1}{16}$ of each HBM is dedicated for caching in a 4-chiplet system, with performance gains increasing up to 15% (10% on average) in 16-chiplet systems. Overall, this thesis provides insights into the design and optimization of multi-chiplet architectures, paving the way for scalable and efficient systems in the post-Moore's Law era.

Keywords: Chiplets, Non-Uniform Memory Access, Caching, Migration

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List of Publications

Appended publications

This thesis is based on the following publications:

[A] Neethu Bal Mallya, Panagiotis Strikos, Bhavishya Goel, Ahsen Ejaz, and Ioannis Sourdis
"A Performance Analysis of Chiplet-Based Systems"
To appear in 2025 Design, Automation and Test in Europe Conference and Exhibition (DATE), Lyon, France, Mar 2025.

[B] Neethu Bal Mallya, Bhavishya Goel, and Ioannis Sourdis
"MEMPLEX: A Multi-Chiplet NUMA Architecture with Data Replication and Migration"
Under review.

Contents

Chapter 1

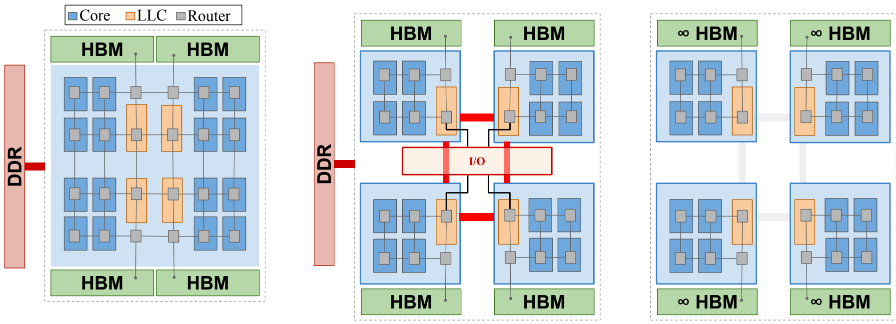
Introduction

In the multicore era, performance scaling effectively relies on integrating more resources onto a chip. The shift occurred because Dennard scaling, which had enabled frequency scaling, was constrained by power limitations. The deceleration of Moore's Law has intensified technology scaling challenges. Large monolithic chips suffer from low yields and high costs, making them less viable. Consequently, building chips from multiple smaller chiplets offers a better cost-effective alternative, providing higher yields and better scalability [1–4].

3D-stacking technology initially involved stacking memory chips, such as High Bandwidth Memory (HBM), and placing them closer to processing die like GPUs [5] or vector engines [6]. This technology has evolved beyond simple memory stacking to enable more complex and efficient chiplet systems, as exemplified in AMD's EPYC and RYZEN architectures [1]. However, despite offering significant advantages in yield and cost, multi-chiplet systems introduce new performance challenges. The size of multi-chiplet systems often necessitates multiple non-uniform access memory (NUMA) nodes, leading to performance bottlenecks that arise from varying memory access latencies. For instance, early AMD EPYC and RYZEN chips connected CPU chiplets to DRAM via a single I/O die, resulting in access latencies that varied by tens of nanoseconds, depending on the DRAM controller being accessed [1]. Multi-chiplet systems have advanced, incorporating more chiplets and complex memory systems, such as in AMD's MI300 [7] and Intel's Sapphire Rapids [8], but the NUMA-related inefficiencies remain a persistent challenge.

This thesis studies the overheads of remote memory access and inter-chiplet communication in multi-chiplet systems and evaluates their impact on system performance compared to traditional monolithic systems. It also proposes solutions to mitigate these challenges in these increasingly prevalent architectures.

The remainder of the introductory chapter is structured as follows: Section 1.1 outlines the problem statement, followed by a discussion of the thesis objectives and contributions in Section 1.2.



(a) Monolithic System (b) Multi-chiplet System (c) Ideal System

Figure 1.1: Hypothetical System Configurations with 16 cores: Monolithic *vs.* Multi-Chiplet *vs.* Ideal

1.1 Problem Statement

In recent years, chiplet architectures have gained attention as a solution to the scaling challenges of traditional monolithic architectures. Multi-chiplet systems offer a low-cost alternative, delivering higher yields and better scalability compared to their monolithic counterparts [1–4]. However, the benefits of chiplet-based systems come with trade-offs, particularly in terms of performance overheads related to remote memory access and inter-chiplet communication.

Without loss of generality and to better understand the trade-offs, Figure 1.1 illustrates typical system configurations for a 16-core multicore architecture, comparing three design paradigms: (a) a monolithic system, (b) a multi-chiplet system, and (c) a hypothetical ideal system. The chip organization and floorplan, inspired by the AMD Zen families [9, 10], consists of compute tiles composed of the core with private L1 and L2 caches and a shared Last-Level Cache (LLC) partitioned in slices. The memory hierarchy incorporates HBM and external DDR DRAM interfaces, with all components interconnected via a Network-on-Chip (NoC). In the monolithic chip, a 2D mesh NoC is employed, with the LLC slices located in the middle columns of the 2D mesh and the HBM and external DDR interfaces placed at the chip's edges. The multi-chiplet system divides the architecture into several chiplets, each containing a subset of tiles, one or more LLC slices, and HBM interfaces, while a separate I/O chiplet manages the external DDR access. Each chiplet utilizes a 2D mesh NoC topology, and inter-chiplet communication is constrained by the chiplet's microbumps budget, as detailed in prior work [11,12]. The high latency and limited bandwidth of the inter-chiplet communication add extra latency compared to the monolithic system. The ideal system serves as a theoretical baseline, assuming infinite capacity in the local HBMs, thus eliminating remote memory accesses and the need for communication across chiplet boundaries.

Remote memory accesses in a NUMA environment increase access latency, while the constrained bandwidth of inter-chiplet links exacerbates communication delays. Consequently, multi-chiplet systems face two primary challenges that limit their performance: (i) NUMA inefficiencies impacting the remote accesses and (ii) inter-chiplet communication overheads. These challenges form the core of this thesis and are discussed in detail in the following sections.

1.1.1 Cost-Performance Trade-offs of Multi-Chiplet Systems Compared to Monolithic Systems

Although the economic effectiveness of multi-chiplet chips has been thoroughly analyzed [2], there is a notable gap in understanding the performance implications of multi-chiplet architectures compared to monolithic designs. Multi-chiplet systems exhibit performance overheads due to the inherent non-uniform memory architecture and inter-chiplet communication. These overheads may negate the yield and cost advantages, making it crucial to evaluate the impact systematically.

This thesis aims to comprehensively evaluate the performance overheads of multi-chiplet systems relative to monolithic architectures and analyze the associated cost-performance trade-offs. The findings will provide valuable insights into the feasibility of multi-chiplet architectures as a scalable and cost-effective alternative to monolithic chips while identifying avenues to mitigate their performance limitations.

1.1.2 NUMA Challenges in Multi-Chiplet Systems

While software can optimize data placement in multi-chiplet NUMA systems, there are currently no hardware mechanisms to improve data placement in DRAM distributed across chiplet nodes at runtime. Our experiments reveal that even with NUMA-aware memory allocation—where data is placed in the closest available memory node relative to the processing node—system performance remains significantly lower than that of an ideal system, which always retrieves data from its local memory node.

While the monolithic system also lags behind the ideal in terms of system performance, the chiplet-based system experiences an even greater slowdown, emphasizing a significant performance gap that must be addressed. This gap arises from the latency and bandwidth penalties associated with remote memory requests, which are inherent to multi-chiplet NUMA architectures. These inefficiencies pose a significant barrier to achieving optimal performance in such systems. Therefore, there is a pressing need for mechanisms that mitigate the performance impact of remote memory accesses.

Together, these two problems—understanding the cost-performance trade-offs of multi-chiplet architectures and addressing NUMA-related inefficiencies—constitute the primary focus of this thesis.

1.2 Thesis Objectives and Contributions

The overarching goal of this thesis is to investigate and optimize the performance of multi-chiplet systems by addressing their inherent NUMA challenges. This involves exploring methods to reduce performance overheads and improve memory access efficiency, specifically by evaluating the impact of different architectural choices and proposing memory system techniques to overcome the existing performance bottlenecks. Below, we outline the specific objectives that drive this thesis and describe the approach taken to achieve these objectives, along with key contributions of the thesis.

1.2.1 Evaluating Cost-Performance Trade-offs of Multi-Chiplet Systems Compared to Monolithic Systems

Objective: The first objective of the thesis is to analyze the performance overheads of chiplet-based systems and evaluate their cost-performance trade-offs compared to monolithic chips. This includes a detailed examination of key design parameters that influence both system performance and cost.

Related Work: The benefits of multi-chiplet systems, particularly in terms of yield and cost, have been thoroughly analyzed in the prior studies [2]. However, these systems are not without performance penalties, primarily due to the need for multiple NUMA nodes and the associated communication latencies. For example, early AMD EPYC and Ryzen processors connected their CPU chiplets to DRAM through a single I/O die, leading to varying access latencies depending on the DRAM controller being accessed [1]. These latencies, often differing by tens of nanoseconds, highlight the inefficiencies in memory access that multi-chiplet systems must overcome.

Thesis Approach: This thesis focuses on evaluating the cost-performance trade-offs of multi-chiplet systems by examining various design parameters, including (i) system size, (ii) chiplet size, (iii) LLC organization, (iv) NoC datapath width, and (v) silicon interposer type (passive vs. active). By analyzing these parameters, the thesis aims to identify configurations that balance performance and cost in multi-chiplet systems. To achieve this, a microarchitectural simulation setup was developed to model large-scale chiplet architectures, and the key design choices and technological factors influencing performance were examined. The performance overheads of chiplet-based systems are evaluated across different design alternatives, with insights drawn from several interconnect and memory system metrics. To complement this analysis, the thesis evaluates the associated costs using the Feng-Ma chiplet actuary model [2], incorporating the parameters of the evaluated systems in our study.

Thesis Contributions: In line with the objective, **Paper A** examined the performance overheads of multi-chiplet systems, which had not been thoroughly analyzed in prior studies, and made the following contributions:

- Developed a microarchitectural simulation setup to model large-scale chiplet-based architectures, with detailed models of their memory system and interconnection networks, capable of simulating about a billion instructions per hour.
- Examined technological factors influencing inter-chiplet link delays, microbump budgets, and the yield and cost of chiplet-based versus monolithic chips, confirming that chiplet-based designs can reduce recurring engineering costs by nearly half.
- Conducted the first systematic performance analysis of chiplet-based systems against monolithic chips, showing that chiplet-based systems

suffer a performance loss of about one-third compared to monolithic. Compared to monolithic, chiplet-based systems:

  - Achieve only 43% to 75% of monolithic performance, averaging 58%, while an ideal system, where LLC misses always go to the closest HBM, is 24% faster than monolithic.
  - Suffer a 40% higher average memory access time, contributing significantly to the performance gap.
  - Experience 3.7× longer average packet latency due to inter-chiplet communication overheads.

- Investigated design parameters such as system size, chiplet size, LLC organization, NoC datapath width, and silicon interposer type, and analyzed their impact on system performance. The key observations include:

  - As the system size increases, the performance gap between monolithic and multi-chiplet systems remains relatively stable.
  - For a constant system size, the performance reduces as the system is disintegrated into more, smaller chiplets.
  - Private LLC improves the system performance by about 30% over Sliced LLC in multi-chiplet systems.
  - Wider intra-chiplet NoC links improve the network throughput, reduce average memory access time, and therefore improve overall system performance.
  - Active interposers effectively recover most of the performance overhead seen in chiplet-based systems with passive interposers but at a 61% higher cost.

Appendix A provide additional results on the cost analysis of various design options for chiplet-based systems, which were not included in **Paper A** due to space constraints. This supplementary material offers further insights and supports the findings presented in **Paper A**.

1.2.2 Mitigating NUMA Challenges in Multi-Chiplet Systems

Objective: The second objective of this thesis is to alleviate this performance degradation in NUMA-based multi-chiplet architectures and bridge the performance gap to ideal systems. The **key insights** behind this work are:

- *Remote memory access in multi-chiplet NUMA systems leads to significant performance degradation due to the high latency and limited bandwidth associated with accessing data located on remote memory nodes.*
- *Caching remote data within the DRAM cache of the local HBM node enhances data locality, thereby reducing the frequency of remote memory accesses.*
- *Migrating data upon the DRAM cache eviction ensures that the most frequently accessed data stays close to the processing chiplet.*

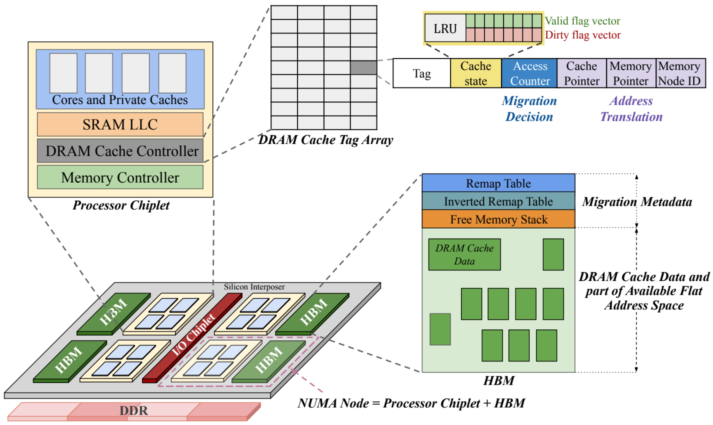


Figure 1.2: MEMPLEX System Overview

Related Work: The performance of multi-chiplet systems is primarily limited by two key challenges: the NUMA-related inefficiencies in memory access and the overheads associated with inter-chiplet communication.

NUMA-related inefficiencies have long been a challenge in computing systems. Early solutions, such as Cache Only Memory Architectures (COMA) [13] and Cache Coherent NUMA (CC-NUMA) [14] memory systems, enhanced the performance of NUMA multi-socket machines by replicating (caching) and/or migrating data closer to the processor chip. More recently, hybrid memory systems, which combine smaller High Bandwidth Memory (HBM) with larger, lower bandwidth external DRAM, have adopted similar techniques to reduce memory access latencies. These hybrid systems use HBM as part of a flat address space, relying on Operating System (OS) support [15] or hardware migration mechanisms [16–21] to place data closer to the processing chiplet, or they use the entire HBM as a DRAM cache for external DDR memory, often wasting valuable main memory capacity [22–31], or employ a combination of both [32].

Another performance overhead in multi-chiplet chips, which exacerbates NUMA effects, stems from the increased latency and bandwidth constraints in inter-chiplet communication. The necessity of traversing longer links through microbumps and silicon interposers introduces additional latency. Moreover, the chiplet size and density limitations restrict the number of available microbumps, capping the available off-chiplet bandwidth and further impacting performance.

Thesis Approach: To address these challenges, this thesis proposes MEMPLEX, a novel memory system designed for multi-chiplet architectures, which incorporates the above insights and techniques to minimize the performance degradation in NUMA-based multi-chiplet architectures. MEMPLEX is a novel memory system that enhances data locality by replicating and migrating data across memory nodes in multi-chiplet systems. MEMPLEX is tailored for chiplet-based architectures, comprising multiple processor chiplets and HBMs integrated on a silicon interposer as well as external DDR memory accessed via an IO chiplet, as illustrated in Figure 1.2. MEMPLEX increases the number of accesses to the closest memory node for each processing chiplet, thereby minimizing remote memory requests. By leveraging a small fraction of each HBM node as a DRAM cache and intelligently deciding whether to migrate data upon eviction, MEMPLEX reduces average memory access times, improves overall system performance, and provides substantial energy savings.

Thesis Contributions: In line with the objective, **Paper B** proposed memory system enhancements to overcome the existing performance overheads, and made the following contributions:

- Investigated the performance bottlenecks in multi-chiplet NUMA systems, revealing performance losses of 26% and 31% in 4- and 16-chiplet configurations, respectively, compared to an ideal system.
- Proposed MEMPLEX, the first multi-chiplet NUMA architecture that combines replication and migration of data across multiple memory nodes, offering:
  - Most of the HBM capacity as a shared flat address space, unlike designs that use it entirely as a DRAM cache.
  - Superior performance to existing software solutions offering NUMA-aware data placement and designs using HBM exclusively as a cache.
- Evaluated MEMPLEX on multi-programmed workloads from different benchmark suites (detailed in Section 3.4.3) and demonstrated that, compared to a multi-chiplet system with NUMA-aware data placement and no support for DRAM caching or migration, MEMPLEX:
  - Eliminates 80% of remote memory traffic, leading to a 44% decrease in dynamic memory energy consumption in a 4-chiplet system.
  - Achieves up to 7% speedup (5% on average) when $\frac{1}{16}$ of each HBM is dedicated for caching in a 4-chiplet system, with performance gains increasing up to 15% (10% on average) in 16-chiplet systems.

1.3 Thesis Outline

The remainder of this thesis is organized as follows. Chapter 2 presents **Paper A**, "*A Performance Analysis of Chiplet-based Systems*," which addresses the first problem, and Chapter 3 presents **Paper B**, "*MEMPLEX: A Memory System with Replication and Migration of Data for Multi-Chiplet NUMA Architectures*," which addresses the second problem of the thesis.